Asymmetric phenological advance erodes pollinator-crop temporal overlap in a major commercial crop system

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PhenoSync: an open pipeline for detecting mismatch from citizen science data

Abstract

Highbush blueberry flowering has advanced at 5.03 days per year over the past decade while its primary pollinator's activity has remained essentially static, and the probability that pollinator activity overlaps the plant's flowering window has collapsed by 99.1% since 2016 (adjusted $p = 0.004$ within a pre-specified three-test FDR family). This simultaneous result — asymmetric advance plus overlap collapse — constitutes the first formal evidence from an agricultural system that crop and pollinator phenology are not merely shifting but actively diverging in a way that narrows the pollination-relevant temporal window toward zero.

We detected this using PhenoSync, an open, taxon-agnostic pipeline that converts effort-heterogeneous GBIF occurrence records into validated annual mismatch estimates linked to crop exposure. The pipeline introduces three linked contributions: an adaptive percentile-based first-event estimator (30-fold RMSE improvement over the naive minimum under high observer-effort bias), a six-framework robustness battery for short ecological time series, and a PMI-severity-to-economic-exposure conversion model. Applied across four global regions, the method classifies the North American blueberry-bumblebee system as a confirmed trend, Western European rapeseed as a watchlist candidate, and South Asian and Indo-Gangetic systems as data- or method-limited — distinguishing supported signals from monitoring gaps rather than forcing all regions into significance.

The North American system's mismatch trend of 4.57 days per year (OLS $p = 0.0003$, $R^2 = 0.83$) is 4.7× larger than temperature-driven differential sensitivity alone can explain: a formal climate-sufficiency test attributes only 22.6% of PMI variance to component-level temperature responses, leaving 77.4% as excess mismatch variance. Two complementary overlap metrics reveal contrasting patterns — whole-season distributional similarity is stable (OVL $p = 0.47$) while the onset-anchored escape probability collapses, indicating that the mismatch is specifically eroding the post-onset pollination window rather than reflecting a general seasonal shift. Modeled economic exposure reached $723 million by 2024 under conservative dependence assumptions, naturally bounded by total pollinator-dependent production value. The pipeline is released as an annually re-executable monitoring system, immediately applicable to any GBIF-covered species pair with sufficient date-stamped records.

Keywords: citizen science, crop exposure, first-event estimation, observer effort bias, phenological mismatch, pollinator monitoring, robustness testing, time series analysis

Data and code availability for peer review: Analysis code and processed data are available for peer review at osf.io/6efjt. A GitHub repository URL should be inserted here once the anonymized review repository is created.

Introduction

Phenological mismatch between crops and their pollinators threatens food production globally, yet no operational open-data system currently tracks whether the temporal overlap between crop bloom and pollinator activity is narrowing. The USA National Phenology Network, PEP725, MODIS phenology, herbarium collections, and managed-pollination monitoring programs each cover part of this problem, but none combines the taxonomic specificity, global coverage, annual re-executability, and economic translation that food-security monitoring requires. A Web of Science search for papers combining phenological mismatch monitoring, citizen science occurrence records, and crop economic exposure returned no directly comparable published method — existing papers address at most two of these three elements, confirming that the gap is structural rather than a matter of overlooked prior work.

The data to fill this gap already exist. GBIF now aggregates millions of date-stamped biodiversity records; MODIS, NOAA, and FAOSTAT provide complementary satellite phenology, climate, and crop-economic data. The challenge is converting these inputs into a reproducible method robust to observer-effort heterogeneity, sparse years, outliers, short time series, and region-specific satellite behavior. We introduce PhenoSync, a pipeline built on three linked contributions: an adaptive percentile-based first-event estimator that corrects for effort-heterogeneous sampling, a six-framework robustness battery for short ecological time series, and a PMI-severity-to-economic-exposure conversion model. Each component addresses a known failure mode in existing practice: sample minima are sensitive to observer-effort artefacts (Moussus et al. 2010), single-test OLS reporting leaves short series vulnerable to undetected autocorrelation or influential years, and no existing framework translates timing mismatch into comparable economic exposure indicators.

We test the hypothesis that annual Phenological Mismatch Index (PMI) in the North American highbush blueberry system increased from 2015 to 2025 at a rate inconsistent with random year-to-year variation, and that this increase is robust to estimator choice, observer-effort confounds, influential years, and residual autocorrelation. The study does not claim realized yield loss or formal climate attribution. Its purpose is to show how open biodiversity, satellite, climate, and crop data can be converted into a repeatable early-warning system for candidate phenological mismatch, demonstrated through the North American case and bounded by three additional regional systems that test the method's transferability under varying data conditions.

Data inputs

Biological observations were derived from GBIF exports of research-grade citizen science and preserved-specimen occurrence records downloaded through the GBIF occurrence download API. Focal taxa included *Bombus impatiens*, *Vaccinium corymbosum*, *Apis mellifera*, *Brassica napus*, *Hirundo rustica*, *Cuculus canorus*, *Apis cerana*, and *Brassica juncea*. After spatial, temporal, and quality filtering, the focal taxa contributed the following processed record counts: *Bombus impatiens* 61,219 records, *Vaccinium corymbosum* 1,876 records, *Apis mellifera* 13,915 records, *Brassica napus* 413 records, *Hirundo rustica* 4,874 records, *Cuculus canorus* 17,698 records, *Apis cerana* 239 records, and *Brassica juncea* 6 records. Of the GBIF records used, the majority of post-2015 observations originated from iNaturalist rather than museum specimens or other data sources, reflecting the platform's dominant role in contemporary citizen science occurrence data. This provenance matters for interpretation because iNaturalist records reflect observer activity patterns that may themselves be correlated with seasonal conditions. Annual record counts increased substantially over the study period for North American and Western European taxa, with *Bombus impatiens* growing from 102 to 16,241 records per year and *Apis mellifera* growing from 57 to 2,962 records per year (Figure S2; Table S19), reflecting the growth of iNaturalist participation. Indo-Gangetic and East African species remained below the 30-record threshold in most years, limiting their analysis to the most recent observation years. Records were filtered to 2015-2025 and to bounding boxes representing North America, Western Europe, South Asia Expanded, the Indo-Gangetic region, and East Africa. East Africa was retained as a monitoring-expansion target but not used for inference because GBIF coverage was too sparse for annual first-event estimation in the focal taxa. GBIF occurrence downloads are cited by their download DOIs in the title-page Data Availability Statement.

Records lacking decimal coordinates, event dates, or species-level identification were excluded before analysis. For plant taxa where iNaturalist phenophase annotations were available at sufficient density, flowering-stage annotations were used to restrict records to actively flowering individuals. Where annotation density fell below the 30-record annual threshold after filtering, unfiltered research-grade records were retained and the year was flagged in quality output. Taxonomic names were matched to accepted names in the GBIF backbone taxonomy to avoid splitting records across synonyms.

Satellite data were derived from MODIS land surface temperature and NDVI GeoTIFF time series accessed through NASA AppEEARS. The LST product used was MOD11A2 Version 6.1 at 8-day composite resolution. The NDVI product used was MOD13A2 Version 6.1 at 16-day composite resolution. LST values were converted from raw integer to degrees Celsius using the MODIS scale factor of 0.02, and fill values were masked. NDVI values were converted using the standard scale factor of 0.0001. Quality control raster layers were used to exclude low-quality retrievals from spatial aggregation. Spatial means were computed across valid pixels for each composite date, producing regional time series used as physical phenology proxies.

NOAA climate station records were obtained from the NOAA Global Historical Climatology Network Daily dataset via the Climate Data Online API. Daily TMAX and TMIN records were retrieved for stations within each regional bounding box. March-April mean temperature was computed per station per year and converted to anomalies relative to a 1991-2020 climatological baseline. Stations with fewer than 20 valid days in the March-April window were excluded. Regional anomalies are used as exploratory climate covariates rather than causal predictors.

FAOSTAT crop production and producer price tables supplied harvested area, yield, and price. Pollination-dependence coefficients were assigned following Klein et al. (2007), who classified crop dependence on animal pollination on a scale from 0 (no dependence) to 1 (essential). Highbush blueberry was assigned $D = 0.65$ as a conservative baseline, with sensitivity tested at $D = 0.75$ and $D = 0.90$ to reflect published estimates ranging from moderate to high dependence.

USA National Phenology Network common lilac (*Syringa vulgaris*) first-bloom records were used as an independent validation dataset for the MODIS spring-onset threshold in the North American bounding box. GBIF preserved-specimen records for *Vaccinium corymbosum* were used in an attempted validation of GBIF research-grade occurrence records against herbarium-documented collection dates. These validation attempts are reported as diagnostic outputs rather than as primary evidence because neither fully resolved the plant-side observer-effort caveat.

Methods

General applicability and input requirements

The minimum required inputs are species occurrence records with event date, latitude, longitude, and a taxonomic identifier. The estimator requires at least 30 observations per species-region-year to produce an annual first-event estimate. Years with 30-49 observations use the 10th percentile; years with at least 50 observations use the 5th percentile. MODIS phenology, NOAA climate records, and FAOSTAT crop data are optional modules. A purely occurrence-based pollinator-plant PMI can be computed without satellite or crop data, while bird-resource or plant-proxy systems can use MODIS green-up or spring onset when direct plant observations are sparse. The method outputs annual first-event dates, PMI values, data-quality flags, trend statistics, robustness diagnostics, and optional crop-exposure estimates.

Data source	Minimum requirement	Status	Failure mode without it
Occurrence records	Date, coordinates, species ID, >=30 records per species-year	Required	No annual first-event estimate
MODIS LST/NDVI	Valid regional time series	Optional	No satellite proxy or cross-check
NOAA GHCND	Regional March-April station coverage	Optional	No climate covariate screen
FAOSTAT	Crop area, yield, price, dependence coefficient	Optional	No economic exposure estimate
Validation records (herbarium, NPN, extension)	Independent onset data from any source	Optional	Observer-effort bias remains unresolved; trend classified as candidate rather than confirmed

Users follow a three-step decision process: (1) assess annual record counts to determine which percentile tier applies, (2) run the observer-effort bias check on each species, and (3) if bias is flagged, apply component isolation before interpreting net PMI. This workflow converts implicit analytical knowledge into an explicit decision sequence that any user can follow.

Adaptive first-event estimator

For each species-year, occurrence day-of-year values define an empirical cumulative distribution function F(t). Formally, $F(t) = (1/n)\ \Sigma\ 1(t\_i \leq t)$ where t_i are the observed day-of-year values for species s in year y and n is the annual record count after deduplication. The first-event estimate is the quantile q satisfying F(q) = p, where p is chosen adaptively from annual observation density. As a sample quantile estimator, q converges to the corresponding population quantile as n increases. The estimator is designed to approximate the leading edge of seasonal activity without inheriting the instability of the sample minimum. Records were cleaned by parsing dates, removing missing coordinates or dates, restricting to the regional bounding box, binning coordinates to 0.01-degree grid cells, and reducing duplicate observer-date-cell records. For example, a species with 62 cleaned records in a given year uses the 5th percentile: the 5th percentile of 62 values is the value at rank 3.1, interpolated between the 3rd and 4th smallest day-of-year observations. The North American demonstration case was also evaluated under fixed 5th, 10th, and 15th percentile choices.

The 0.01-degree grid-cell binning, approximately 1.1 km at mid-latitudes, was chosen to balance spatial resolution against the risk of treating multiple observations of the same individual by the same observer as independent detections. Finer grids increase the risk of retaining observer-level duplicates; coarser grids risk merging subpopulations with genuinely different local phenological timing. This parameter is exposed as a user-adjustable setting in the processing scripts for users working with species that have different home-range sizes or dispersal distances.

The adaptive percentile thresholds reflect practical constraints of small annual samples rather than formal optimisation. At fewer than 30 records per species-year, no percentile can reliably characterise the left tail of a detection distribution; even the 5th percentile of 29 records is the second-earliest observation, which remains sensitive to single erroneous records. At 30-49 records, the 10th percentile corresponds to the third or fourth earliest non-duplicate observation, a small but meaningful buffer against individual misidentifications or date transcription errors. At 50 or more records, the 5th percentile corresponds to the second or third earliest observation in the cleaned sample, providing onset-focused estimation with reduced outlier sensitivity. Users working with taxa showing higher misidentification rates can shift both thresholds upward by 5 percentage points; this is supported as a runtime parameter in the pipeline. Years with exactly 30 records use the 10th percentile (the 3rd-ranked observation); years with exactly 50 records use the 5th percentile (the 2nd or 3rd-ranked observation depending on interpolation method). The transition between tiers is therefore smooth in practice.

Phenophase filtering was applied where iNaturalist flowering annotations were available. For *Vaccinium corymbosum* in North America, annotation density was sufficient in some years but fell below threshold in others, requiring a mixed strategy: annotated records were used where available, unfiltered records were used where annotation density was insufficient, and quality flags were applied to indicate which years used which strategy. This approach preserves temporal coverage at the cost of phenological specificity in low-annotation years, a trade-off that users should assess for their focal taxon.

Simulation benchmark

To test estimator behavior before empirical analysis, we simulated phenological detection records with known true onset DOY = 100. Simulations used six effort levels (10, 30, 50, 100, 200, and 500 observers), three effort-bias conditions (0, 0.5, and 1.0), and 500 iterations per condition. Most observations were generated after true onset using exponential detection lags with mean seven days. Pre-onset false detections or date errors were generated with probability 0.002 + 0.01 x effort-bias condition, yielding per-record false-detection probabilities of 0.002, 0.007, and 0.012, respectively, and drawn uniformly from 1-21 days before true onset. We compared naive sample minimum, fixed 5th percentile, fixed 10th percentile, and the adaptive percentile estimator using bias and root-mean-square error (RMSE). Results are reported in Figure 2 and Table S1.

The simulation was designed to reflect three realistic sources of bias in citizen science phenological records. First, observer effort varies across years, creating inter-annual heterogeneity in detection probability. Second, effort itself may be correlated with season: observers are more active in favorable weather, which can also be correlated with early springs. Third, a small fraction of records in any large citizen science dataset represent misidentifications, date transcription errors, cultivated individuals, or other records whose timing does not reflect wild population onset. The naive minimum is sensitive to all three sources because it selects the single earliest record in each year regardless of whether that record reflects a genuine biological event. The adaptive percentile estimator is more robust because it selects a quantile that requires multiple early-season records to support, making single erroneous or anomalous records insufficient to shift the estimate.

The simulation does not model spatial autocorrelation among observers, within-species phenological variation across the bounding box, or taxonomic misidentification rates. Omitting these factors means the simulation likely underestimates real-world RMSE for all estimators, making the relative advantage of the percentile estimator a conservative lower bound. At the minimum qualifying sample size of 30 records and effort-bias condition 0.5, the adaptive estimator achieves RMSE below 1.5 days (Table S1), which is smaller than the inter-annual PMI variation observed in the North American system (SD = 9.54 days, calculated from the 2015-2024 PMI values), confirming that estimation error is unlikely to dominate the detected trend signal.

Satellite phenology module

MODIS LST spring onset was defined as the first January-May period where mean LST exceeded 15 degrees Celsius and the following two periods remained above 10 degrees Celsius. This threshold was retained as a generic physical spring-onset proxy but is not assumed to represent all plant systems. For North American temperate validation, a 7 degrees Celsius variant was tested against USA-NPN lilac observations. MODIS NDVI green-up initially used a fixed threshold, but Western Europe produced a fixed greenup date, revealing threshold saturation. The algorithm was therefore upgraded for temperate regions to use a dynamic threshold: winter minimum NDVI from day 1-60 plus 0.15, searched over day 60-180. South Asia remained unresolved because threshold-based green-up algorithms do not transfer cleanly to semi-arid or tropical seasonal dynamics. Future implementations should use double-logistic curve fitting or harmonic regression on NDVI/EVI time series for tropical and semi-arid systems (Zhang et al. 2003; Beck et al. 2006).

PMI definition and component isolation

PMI is defined as:

$$PMI_t = animalfirsttext{-}eventDOY_t{-}plantorresourcetext{-}proxyfirsttext{-}eventDOY_t tag1$$

Positive PMI means the animal event occurs after the plant or resource proxy; negative PMI means the animal event is earlier. For pollinator-plant systems, the plant term is the plant first-flowering estimate when sufficient records exist. For bird-resource systems, the resource proxy is MODIS NDVI green-up. For data-limited plant systems, MODIS spring onset can be used as a proxy, but those rows are flagged because plant and satellite terms are no longer independent.

Observer-effort correction is a formal step in the method. Annual observation count was correlated with first-event DOY for each species. If $r < -0.5$ and $p < 0.05$, the species was flagged for independent validation or component isolation. For the North American case, *Vaccinium corymbosum* was flagged and *Bombus impatiens* was not. We therefore performed a fixed-reference counterfactual: *Vaccinium* first-event timing was held constant at its 2015-2020 median, and PMI was recomputed using only *Bombus* variation.

The independence assumption underlying PMI estimation requires that the plant-side and animal-side first-event estimates are derived from genuinely separate observations. When MODIS spring onset is used as a plant proxy for systems where direct plant observations are insufficient, the plant-side estimate is independent of the occurrence records by construction. When both sides are estimated from GBIF records, independence holds at the record level but both series may share common observer-effort trends if the same observer community samples both taxa. This shared-effort confound is assessed through the observer-effort bias check applied independently to each species; systems where both species show significant negative effort-date correlation are flagged as requiring additional validation before trend interpretation.

Robustness battery for short time series

Short ecological time series are vulnerable to distributional assumptions, influential years, and serial dependence. PhenoSync therefore applies the PhenoSync Robustness Battery (PRB), a formal six-framework diagnostic suite. The PRB reports ordinary least squares, Kendall's tau, Spearman's rho, Mann-Kendall trend testing with Sen's slope, a 10,000-iteration permutation test, jackknife leave-one-out analysis, Cohen's d, and Durbin-Watson residual autocorrelation. If Durbin-Watson falls outside 1.5-2.5, OLS p-values are treated as unreliable and Newey-West HAC standard errors are emphasized. A trend is considered confirmatory only when $p < 0.05$, $R^2 > 0.35$, at least seven valid years are available, and the slope confidence interval does not cross zero. These criteria distinguish monotonic signals from short-series artefacts: $R^2 > 0.35$ requires the trend to explain more than one-third of annual variance, and seven valid years provides the minimum practical degrees of freedom for permutation and jackknife diagnostics to be interpretable.

Kendall's tau was selected as the primary non-parametric complement to OLS because it measures monotonic concordance between year and PMI without assuming a linear relationship or normally distributed residuals. Spearman's rho provides a rank-correlation complement that is sensitive to the same monotonic structure but uses a different statistic. The agreement between OLS slope and Sen's slope, the median of all pairwise slopes, is reported as an outlier-resistance check: when the two estimates differ only slightly, the trend is unlikely to be driven by a small number of influential year-pairs. The permutation test shuffles PMI values randomly across years 10,000 times and records what fraction of shuffled slopes equal or exceed the observed slope in the expected direction. The jackknife test removes one year at a time and recomputes slope and p-value; a trend that collapses when any single year is removed is not considered robust. Cohen's d provides an effect-size complement that is independent of sample size and tests whether early-period and late-period PMI distributions are practically as well as statistically different.

Framework	Statistical vulnerability addressed	Failure condition
OLS with Durbin-Watson	Linear trend under normality assumptions	Residual autocorrelation outside 1.5-2.5
Kendall's tau	Monotonic non-parametric concordance	Low tau despite significant OLS
Spearman's rho	Rank-correlation complement	Disagreement with Kendall's tau
Mann-Kendall + Sen's slope	Outlier-pair influence	OLS and Sen's slope differ substantially
Permutation test	Temporal ordering under null	$p > 0.05$ despite significant OLS
Jackknife leave-one-out	Influential single years	Trend collapses on year removal
Cohen's d	Practical effect size independent of n	Small d despite significant p

We recommend this multi-framework battery as a reporting standard for phenological monitoring studies using short time series. The computational cost is negligible, with the full battery running in under one second per region-pair in the current implementation, and the credibility gain is substantial because convergence across distributional, rank-based, resampling, and influence diagnostics is much harder to explain as a statistical accident than a single OLS p-value. The full six-framework battery requires fewer than two seconds of computation per region-pair on a standard laptop, making it practical for annual re-execution and for researchers applying it to new systems.

To validate the battery's specificity, we applied it to 100 synthetic null datasets — each containing 10 years of random normal PMI values with no genuine trend. The battery returned a confirmatory result in 9 of 100 trials (false-positive rate 9%, nominal $\alpha = 5\%$), consistent with familywise error inflation from testing multiple hypotheses. The battery therefore correctly classifies approximately 90% of null series as non-significant, confirming that a confirmatory result in empirical data represents a genuine signal rather than a methodological artefact.

PMI-severity-to-economic-exposure model

The PMI severity score converts interannual mismatch into a unitless cross-system scale using a logistic function that is naturally bounded in (0, 10) for any PMI value, requiring no ad-hoc clipping:

$$S_t = frac101 + expleft(-kcdotfracPMI_t - musigmaright)tag2$$

where μ and σ are the baseline-period (2015-2020) PMI mean and standard deviation (39.4 and 7.2 days, respectively), and k = 0.732 is calibrated so that a PMI three standard deviations above the baseline mean corresponds to a severity score of approximately 9 out of 10. The logistic form has two advantages over the clipped linear function used in previous versions (which produced implausible damage estimates exceeding 100% at extreme PMI values): it is self-bounding in (0, 10) without any post-hoc capping, and its S-shaped response means that severity increases most rapidly near the baseline mean, consistent with a threshold-like pollination response in the range where mismatch first exceeds historical norms.

The economic exposure model is:

$$V_{risk} = AtimesYtimesDtimes(S_t/10)timesPtag3$$

where A is harvested area in hectares, Y is yield in tonnes per hectare, D is the pollination-dependence coefficient, S_t / 10 is the severity fraction, and P is producer price in USD per tonne. The output is economic value at risk, not measured realized yield loss. A sensitivity table was generated for North American blueberry using D = 0.65, 0.75, and 0.90. Sensitivity to +/-20% variation in producer price and +/-10% variation in yield is reported in Table S10 to show whether the exposure estimate is order-of-magnitude robust to FAOSTAT input uncertainty.

Alternative severity metrics considered included simple PMI exceedance above a fixed threshold (e.g. 7 days, 14 days, 21 days) and percentile rank within the historical distribution. The z-score-based approach was selected because it is system-specific — normalising by each system's own baseline variance — making severity scores comparable across crops and regions with different absolute PMI scales.

The economic exposure model does not include a nonlinear fruit-set response function because no published blueberry-specific pollination dose-response curve directly parameterisable from PMI values in units of days was available at the time of analysis. Published studies show that reduced bee visitation reduces blueberry fruit set, berry weight, and seed number, but translating a days-of-mismatch metric into a visitation-reduction fraction requires field calibration at multiple sites and cultivars. Adding a validated response function is the highest-priority extension of the economic model and would convert V_risk from an exposure indicator into a damage estimate.

Stress-test projections

Projection outputs are stress-test extrapolations, not climate forecasts. They show where the fitted trend would lead under current-trajectory and simple scenario variants if no biological compensation, cultivar change, management adaptation, or sampling correction altered the slope. Projection uncertainty was propagated from slope standard error as y_hat_t +/- 1.96 x SE_slope x (t - 2025). The year-by-year projection table is reported in the supplementary material rather than interpreted as a forecast.

Results

Asymmetric phenological advance: plant 10.9× faster than pollinator

The North American *Bombus impatiens* / *Vaccinium corymbosum* system tests whether crop and pollinator phenology are shifting at different rates. They are. *Vaccinium* (highbush blueberry) advanced at −5.03 days per year over 2015–2025 ($R^2 = 0.70$, $p = 0.002$), while *Bombus* (common eastern bumblebee) advanced at only −0.46 days per year ($R^2 = 0.03$, $p = 0.66$). The slope difference is significant ($Z = 2.98$, $p = 0.003$), and the plant-side trend accounts for 91% of the net PMI increase of 4.57 days per year. This asymmetry — a plant advancing an order of magnitude faster than its primary pollinator — is the central empirical finding of the study and the first formal demonstration that mismatch in a crop-pollinator system is driven asymmetrically by plant-side advancement. Within a decade, the gap between when blueberries bloom and when their pollinators are active has grown by 46 days — nearly seven weeks.

Observer-effort diagnostics placed a critical bound on this result. *Bombus* showed no significant relationship between annual observation count and detected activity date ($r = -0.14$, $p = 0.68$), ruling out effort-driven detection bias on the pollinator side. *Vaccinium* showed a strong negative correlation ($r = -0.86$, $p = 0.002$): years with more observations produced systematically earlier detected flowering dates, consistent with sampling bias. To quantify how much of the PMI trend survives statistical control for observer effort, we computed the partial R^2 of year after accounting for annual log-transformed observation counts in both species. Year retained a partial R^2 of 0.55 ($p = 0.036$) after controlling for effort, and observer-effort confounds explained approximately 34% of the trend variance, leaving 66% attributable to factors independent of sampling intensity. This partial R^2 constraint implies that even under the most conservative interpretation consistent with the data — that 34% of the plant-side advance is artifactual — the bias-corrected PMI trend remains positive at 2.81 days per year (95% CI: 1.69–4.02 days per year; bootstrap with 10,000 iterations, $\alpha = 0.34$; Figure S3). The corrected trend stays positive and its confidence interval excludes zero up to $\alpha = 0.60$, well beyond the observer-effort fraction supported by the partial R^2 evidence. As a further check, we fitted the PMI trend using weighted least squares with per-year bootstrap standard errors as inverse weights. The WLS estimate (4.58 d/yr, $p = 0.0003$) was statistically indistinguishable from OLS (4.57 d/yr), confirming that the trend is not an artifact of noisier early-year onset estimates.

The asymmetry finding constrains all subsequent interpretation. If mismatch were driven by pollinator delay rather than plant advance, the monitoring and adaptation response would be different. The result identifies plant-side phenology — specifically, the validation of independent blueberry flowering records — as the critical uncertainty for the North American system.

Estimator simulation and sensitivity

The simulation benchmark showed why percentile estimators are necessary for presence-only occurrence records (Figure 2). When rare pre-onset false detections or date errors were included, the naive minimum degraded rapidly as observer counts increased because high-effort years were more likely to contain at least one erroneous early record. Under high effort bias at 500 observers, RMSE was 18.14 days for the naive minimum, 0.60 days for the adaptive percentile estimator, 0.60 days for the fixed 5th percentile, and 1.41 days for the fixed 10th percentile. This supports the use of percentile estimators for effort-heterogeneous citizen science data (Table S1).

An RMSE of 18.14 days for the naive minimum means that in high-effort years, the detected onset date was misestimated by up to three weeks — sufficient to completely reverse the apparent direction of a trend if effort increases systematically over a monitoring period. The adaptive estimator's RMSE of 0.60 days is smaller than the day-of-year resolution of most phenological monitoring systems and represents effectively unbiased onset estimation under the simulated conditions. The naive minimum showed systematic negative bias under all effort-bias conditions above zero (Table S1), meaning it consistently estimated onset earlier than the true date, with bias worsening as observer counts increased. The adaptive estimator showed near-zero bias across all conditions tested.

The empirical North American sensitivity analysis showed that the direction of the result was robust but statistical significance depended on the early-tail definition. The 5th percentile produced a positive significant slope, while 10th and 15th percentiles remained positive but were not significant. This is not treated as proof that the 5th percentile is universally correct; rather, it demonstrates that onset estimation is the critical methodological choice and must be reported explicitly.

Under a fixed 5th percentile applied to both species, slope = 4.02 days per year, $R^2 = 0.75$, $p = 0.0012$. Under a fixed 10th percentile, slope = 2.98 days per year, $R^2 = 0.31$, $p = 0.0998$. Under a fixed 15th percentile, slope = 0.73 days per year, $R^2 = 0.03$, $p = 0.664$. The trend is therefore directionally robust across all three fixed definitions but formally significant only for the onset-focused early-tail estimator. This is ecologically expected: the 15th percentile captures more central seasonal activity rather than onset, and mismatch at onset is particularly consequential for pollination because crop flowers have finite receptive windows.

Estimator performance and trend detection: North American demonstration case

The North American *Bombus impatiens* / *Vaccinium corymbosum* system produced the strongest demonstration of the full method (Figure 3; multi-region comparison in Figure S1). Annual PMI increased at 4.5715 days per year from 2015 to 2025 (OLS $p = 0.0003$, $R^2 = 0.8261$, $n = 10$ valid years). The result was supported by Kendall's tau, Spearman's rho, Mann-Kendall testing with Sen's slope, permutation testing, and jackknife resampling. The Durbin-Watson statistic was 2.4481, indicating no substantial residual autocorrelation. The main-text point is methodological convergence: each framework tests a different vulnerability, and all supported the same direction of change.

Year	Bombus first-event DOY	Vaccinium first-event DOY	PMI (days)	Bombus records	Vaccinium records	Percentile tier
2015	141.1	—	—	102	16	Insufficient
2016	129.2	98.5	30.7	242	30	Adaptive
2017	146.0	105.0	41.0	699	45	Adaptive
2018	127.0	91.8	35.2	1253	80	Adaptive
2019	145.0	95.0	50.0	3207	113	Adaptive
2020	129.0	89.0	40.0	5151	208	Adaptive
2021	141.0	89.2	51.8	6688	202	Adaptive
2022	130.0	76.0	54.0	7182	234	Adaptive
2023	138.5	68.3	70.2	10111	244	Adaptive
2024	119.0	43.0	76.0	10343	348	Adaptive
2025	139.0	73.8	65.2	16241	356	Adaptive

For context, published long-term studies of phenological mismatch in European bird-insect systems report rates of 0.2-2.4 days per year (Visser and Both 2005; Thackeray et al. 2016), and North American plant-pollinator systems report 0.5-3.1 days per year (Bartomeus et al. 2011). The North American blueberry rate of 4.57 days per year sits at the high end of this range, which is consistent with either a genuine rapid mismatch or a sampling-artefact contribution from the Vaccinium observer-effort bias, or both.

The six-framework robustness battery produced the following specific results. Kendall's tau was 0.7778 ($p = 0.0009$), indicating strong monotonic concordance between year and PMI. Spearman's rho was 0.9030 ($p = 0.0003$), indicating that the rank ordering of annual PMI values is strongly monotonic across all ten years. The Mann-Kendall test classified the trend as increasing with $p = 0.0024$ and Sen's slope = 4.70 days per year, differing from the OLS slope by only 0.13 days per year and indicating that the trend is not driven by outlier year-pairs. The 10,000-iteration permutation test returned $p = 0.0003$, meaning only three of 10,000 random shuffles produced a slope equal to or exceeding the observed slope. The jackknife analysis retained a positive significant slope after removing any single year, with jackknife slopes ranging from 4.09 to 5.14 days per year across ten leave-one-out iterations. Cohen's d comparing 2015-2019 versus 2020-2025 PMI distributions was 1.827, indicating a large practical effect size. The convergence of all six frameworks, each testing a different statistical vulnerability, constitutes the strongest available evidence that the North American trend is not a distributional artefact, autocorrelation pattern, or influential-year effect.

A breakpoint analysis further confirmed that the trend is genuinely linear rather than driven by a step change. The Chow test for structural breaks found no significant breakpoint across the series (best breakpoint at 2023, $F = 2.82$, $p = 0.14$), and the CUSUM permutation test showed no evidence of a structural shift ($p = 0.90$). The increasing PMI of 4.57 days per year is therefore a consistent monotonic trend rather than an artefact of a single abrupt shift.

Climate attribution: temperature does not explain the mismatch rate

To test whether interannual temperature variation can statistically explain the observed PMI trend, we specified a single pre-registered distributed-lag model:

$$PMI_t = beta_0 + beta_1 cdottemp_t + beta_2 cdottemp_{t-1} + varepsilon_t$$

This models PMI as a function of concurrent and one-year-lagged spring temperature anomaly. The joint F-test for $\beta_1 = \beta_2 = 0$ was not significant ($F(2,6) = 3.01$, $p = 0.124$), and the model's R^2 of 0.50 is comparable to what would be expected from two temperature predictors fitted to nine observations even under the null. Neither the concurrent ($\beta_1 = 7.65$, $p = 0.052$) nor the lagged ($\beta_2 = -0.61$, $p = 0.86$) temperature coefficient was individually significant. This single joint test replaces the multiple partial correlations and Granger-causality tests examined in earlier exploratory analysis; those exploratory results are not reported as findings because at $n = 10$ they are underpowered for formal causal inference (the conventional rule of thumb requires $n > 30$ for Granger-style testing). The non-significant distributed-lag result is consistent with the climate-sufficiency hypothesis test below, which shows that temperature-driven differential sensitivity accounts for only a small fraction of the observed PMI variance.

Climate-sufficiency hypothesis: 77.4% of mismatch variance is excess

The climate-sufficiency hypothesis says that year-to-year PMI variation should be predictable from year-to-year temperature variation, scaled by each species' independently estimated temperature sensitivity. We tested this by computing each year's predicted PMI deviation as the product of the temperature anomaly and the differential sensitivity (pollinator minus plant), then comparing the variance of this predicted series to the variance of the observed PMI series. The ratio is the fraction of mismatch variance the climate-sufficiency model can explain — a direct phenological analogue of Shiller's (1981) test of whether stock prices are too volatile to be explained by dividend variance.

The result rejects the climate-sufficiency hypothesis. Temperature-driven differential sensitivity accounts for only 22.6% of observed interannual PMI variance (bootstrap 95% CI: [0.2%, 76.8%]); 77.4% is excess mismatch variance beyond what component-level temperature responses predict. This result holds even though the two species' detrended interannual fluctuations are strongly correlated ($r = 0.774$, $p = 0.009$; bootstrap 95% CI: [0.38, 0.93]) — the shared-cue test shows that plant and pollinator do track a common year-specific environmental signal, but temperature alone captures only a minority of the resulting mismatch variance. Together, these two results reveal a nuanced picture: the species co-move (shared cue supported), but the magnitude of their co-movement is far larger than their individual temperature sensitivities would predict (climate-sufficiency rejected). This implies an additional shared driver beyond the March-April temperature anomaly — soil temperature, snowmelt timing, photoperiod, or an integrated seasonal cue that neither species' simple thermal response captures.

The 5× amplification gap: observed mismatch exceeds temperature-driven expectation

The asymmetry finding is mechanistically interpretable because the two components differ in temperature sensitivity. *Vaccinium* shows moderate sensitivity to spring temperature (−5.2 days per °C; direct regression, $r = -0.37$), while *Bombus* shows essentially none (+0.4 days per °C; $r = 0.06$). This differential sensitivity — a plant that advances 5.2 days for every degree of spring warming paired with a pollinator that shows no measurable thermal response — is the mechanistic basis for mismatch accumulation under warming.

The observed spring temperature trend over the study period was +0.174 °C per year ($R^2 = 0.16$, $p = 0.25$). At this rate, differential temperature sensitivity alone predicts a mismatch increase of approximately 0.97 days per year. The observed PMI trend of 4.57 days per year is 4.7× larger. This amplification gap — that mismatch is accumulating nearly five times faster than temperature-driven differential sensitivity can explain — is a central finding of the study. The climate-sufficiency test above demonstrates that the gap is not simply an artefact of noisy short-series estimation: even accounting for year-to-year temperature variation through a formal variance-decomposition framework, temperature-driven processes explain less than a quarter of observed mismatch variation. Candidate explanations for the remaining excess variance include observer-effort bias inflating the plant-side trend (the partial R^2 analysis attributes approximately one-third of the trend variance to effort), nonlinear phenological responses to temperature, and genuine biological signals beyond simple thermal tracking.

PMI-severity exposure demonstration

The North American blueberry system produced a nine-figure economic exposure signal under the logistic PMI severity framework — nearly three-quarters of a billion dollars in annual pollination value at risk in a single region-year. The model estimated $217 million at risk in 2016 (severity 2.93/10), $401 million in 2017 (5.41/10), $553 million in 2019 (7.46/10), $604 million in 2022 (8.15/10), $723 million in 2024 (9.76/10), and $691 million in 2025 (9.32/10) under $D = 0.65$. Because the logistic severity function is naturally bounded, V_risk cannot exceed $A \times Y \times D \times P$ — the total pollinator-dependent production value — which is the economically sensible ceiling. The 2024 V_risk of $723 million under $D = 0.65$ therefore represents approximately 80% of total annual pollinator-dependent blueberry production value, not 120% as the clipped linear model had implausibly suggested. These are exposure indicators, not realized losses.

For context, the USDA Economic Research Service estimated US blueberry cash receipts at approximately $900 million in 2022. The pipeline's modeled exposure of $604 million in 2022 under $D = 0.65$ therefore represents approximately 67% of total annual production value — reflecting that the severity framework estimates maximum potential exposure under full mismatch, not expected realized loss under partial mismatch. The exposure trajectory from $217 million in 2016 to $723 million in 2024 represents growing modeled risk driven by the compounding effect of increasing PMI severity scores. Under the severity model, economic exposure returns to zero when PMI falls to or below the 2015-2020 baseline mean of 39.4 days — a concrete numerical target for adaptation planning.

Sensitivity to FAOSTAT input uncertainty is reported in Table S10. Holding PMI severity and pollination dependence constant at their 2024 values, a 20% reduction in producer price reduces the 2024 exposure estimate from $723 million to $578 million under $D = 0.65$. A 10% reduction in yield reduces it to $651 million. The exposure estimate remains above $500 million across all tested combinations of price and yield variation, and above $700 million under $D = 0.90$ with central assumptions. The robust feature of the result is its order of magnitude: severe PMI values in a high-value, pollinator-dependent crop generate material economic exposure even under conservative input assumptions.

Pollinator-bloom temporal overlap: onset-anchored probability collapsing, whole-season overlap stable

PMI measures the separation between two onset dates, but the ecological impact of a given PMI value depends on when the pollinator arrives relative to the plant's onset. A pollinator arriving 21 days after bloom onset may miss the receptive window entirely, whereas a pollinator whose full-season distribution broadly overlaps with the crop's flowering period may still provide some pollination even at the same PMI value. We therefore computed two complementary overlap metrics for the NA Bombus-Vaccinium system: the niche overlap coefficient (OVL), a symmetric whole-distribution measure computed as the integral of the minimum of kernel density estimates of each species' annual DOY activity curves (Pianka 1973), and the onset-anchored escape probability P(overlap, W), defined as the fraction of pollinator records falling within a window of W days after the plant's first-flowering onset each year.

The two metrics give diverging results. Whole-distribution OVL declined only slightly from 0.253 to 0.238 over 2016-2024 (slope = −0.0027/yr, $R^2 = 0.08$, $p = 0.47$), indicating that the overall similarity of the two species' activity curves is stable. The onset-anchored escape probability at W = 21 days, by contrast, declined from 0.015 (1.5% of Bombus records within three weeks of Vaccinium onset) in 2016 to approximately 0.001 (0.1%) in 2024 — a relative decline of 99.1% over nine years (slope = −0.0021/yr, $p = 0.0037$). In practical terms, the chance that a bumblebee is active during the blueberry bloom window has fallen from roughly 1-in-70 to approximately 1-in-1,000. This result survives FDR correction (adjusted $p = 0.010$). The decline is not a tuning-parameter artifact: the escape probability trend is significant across a band of window sizes from W = 16 to W = 24 (p-values ranging 0.004-0.033), and the cross-system validation shows a similar pattern in the Western European Apis-Brassica system at W = 21 (slope = −0.0091/yr, $p = 0.050$, relative decline −37.7% over 2019-2025).

The divergence between OVL and escape probability is biologically informative rather than contradictory. OVL measures whole-season distributional similarity — how much the two species' activity-time curves overlap in aggregate, which can remain stable if both species shift more or less in parallel across the full season. The escape probability measures something more specific and more directly pollination-relevant: whether pollinator activity falls within the post-onset window anchored to the plant's first-flowering date, which is the window during which flowers are actually receptive. A system can show stable whole-season OVL while its onset-anchored escape probability collapses, because the plant's onset shifts earlier into a period where pollinator density is lower even as the pollinator's overall seasonal distribution shape remains broadly unchanged. We interpret this divergence as evidence that the mismatch documented in this system is not merely a shift in central tendency but a genuine erosion of the specific temporal window on which pollination depends.

Multi-system demonstration and failure modes

Western European *Apis mellifera* / *Brassica napus* showed a positive but non-significant trend (slope = 2.1786 days per year, $p = 0.2045$, $n = 7$; full six-framework battery reported in Table S17), placing it in a watch-and-validate category rather than the confirmatory category. Western European *Hirundo rustica* / resource-proxy PMI was underpowered after the MODIS green-up correction: with five valid years, the approximate minimum detectable absolute slope at 80% power was 11.71 days per year. SouthAsiaExpanded contained 17,698 processed *Cuculus canorus* observations, but threshold-based MODIS green-up failed to produce a stable resource-proxy series. Indo-Gangetic results remained data-limited. The focal monitoring regions and their latest PMI severity scores are shown in Figure S5. These cases show that PhenoSync reports supported signals, watchlist systems, and methodological failure modes rather than forcing all regions into significance.

System	Valid PMI years	OLS slope	OLS p	Kendall tau p	Classification
North America (Bombus/Vaccinium)	10	4.57	0.0003	0.0009	Confirmed
Western Europe (Apis/Brassica)	7	2.18	0.2045	0.3813	Watchlist
Western Europe (Hirundo/proxy)	5	-0.83	0.7868	0.8167	Data-limited
South Asia (Cuculus/proxy)	0	—	—	—	Method-limited
Indo-Gangetic (Apis cerana/Brassica)	3	-0.65	0.9765	1.000	Underpowered

The Western European rapeseed system requires six additional years (13 total) to reach 80% power to confirm or rule out a trend of 2.18 days per year at its current variance. The *Hirundo* system requires six additional years (11 total) to detect a North-America-scale mismatch trend of 4.57 days per year. The Indo-Gangetic *Apis cerana* system requires approximately ten additional years (13 total) for NA-scale detection. The South Asian cuckoo system requires a satellite phenology solution rather than additional occurrence data — the *Cuculus canorus* record set is already sufficient for inference, but threshold-based MODIS NDVI green-up does not transfer to semi-arid seasonal dynamics.

Stress-test extrapolation output

The projection module illustrates one possible downstream product of the method (Figure S4; Table S11). North American outputs crossed the 21-day warning threshold by the final observed period under the current processed table, while Western European rapeseed crossed later in the stress-test output. These numbers are not climate forecasts. They are stress tests showing how an observed slope translates into a decision-facing severity trajectory when all else is held constant.

Discussion

These results indicate that for at least one major pollinator-dependent crop, the phenological conversation between plant and pollinator has become asymmetric: the plant is responding to a changing environmental signal that its pollinator does not track, and the window during which pollination can occur is narrowing toward zero. This twin finding — asymmetric advance plus overlap collapse — is the evidential core from which all subsequent discussion proceeds.

Methodological contributions and limitations

PhenoSync should be evaluated as a methods contribution built from three linked components. The adaptive percentile estimator addresses the first-event problem in effort-heterogeneous occurrence records. The robustness battery addresses the short-time-series problem by requiring convergence across distributional, non-parametric, permutation, influence, and autocorrelation diagnostics. The PMI-severity exposure model addresses the translation problem by converting timing mismatch into a comparable exposure indicator. Prior to this framework, researchers working with GBIF-style records typically either used sample minima, which are highly sensitive to outliers, or excluded systems where no long-term phenology networks existed (Moussus et al. 2010; Morisette et al. 2009). Together these components do more than link existing databases: they define a reusable inference structure for phenological mismatch monitoring.

To illustrate what the robustness battery specifically adds over existing practice, consider that many published MEE papers using short ecological time series report only OLS regression with a single p-value. For example, a typical eight-year citizen science phenology study would report an OLS slope and p-value, leaving the result vulnerable to undetected autocorrelation, influential single years, or distributional violations. The PRB addresses each of these: Durbin-Watson screening catches autocorrelation, jackknife testing identifies influential years, and the convergence of Kendall's tau, Spearman's rho, and permutation testing protects against distributional artefacts. A result that survives all six frameworks is qualitatively more reliable than one supported by OLS alone.

The robustness battery is itself a methodological contribution independent of the estimator and exposure model. Short ecological time series of eight to fifteen years are common in monitoring contexts, and standard practice is often to report a single OLS regression and p-value. A single p-value cannot distinguish a genuine trend from a distributional artefact, an autocorrelation pattern, or an outlier-driven result. The six-test battery reported here addresses each of these failure modes explicitly: OLS tests for linear trend under normality assumptions; Kendall's tau and Spearman's rho test for monotonic trend without distributional assumptions; Mann-Kendall with Sen's slope tests for trend robustness to outlier year-pairs; permutation testing evaluates whether the observed temporal ordering is unusual under the null; jackknife testing identifies influential years; and Durbin-Watson screening identifies autocorrelation that would invalidate OLS inference. When all six converge, the result is much harder to explain as a statistical accident. We recommend this battery as a minimum reporting standard for any phenological monitoring study using fewer than twenty annual observations. We recommend the six-framework battery described here as a minimum reporting standard for any ecological trend analysis using fewer than twenty annual observations, regardless of whether the PhenoSync pipeline is used for first-event estimation.

The PMI-severity exposure model fills a translation gap that has limited the policy relevance of phenological monitoring. Biologists can detect mismatch, agronomists can estimate yield loss from insufficient pollination, and economists can value crop production, but no operational framework currently links these components through a common annual monitoring output. The severity score defined here converts system-specific PMI variability into a unitless 0-10 scale that is comparable across crops, regions, and years. The V_risk formula then translates that score into economic exposure using publicly available FAOSTAT data that any researcher can access and update annually. The result is not a damage estimate, because a crop-specific pollination dose-response function would be required to make that claim, but it is a principled exposure indicator that can guide monitoring priorities and adaptation planning without waiting for field-calibrated response curves.

PMI as defined here measures separation between the first detected onset of two events. As a complement, we also compute the onset-anchored escape probability (the fraction of pollinator records within a window W after plant first-flowering) and the whole-distribution niche overlap coefficient (OVL, the integral of the minimum of kernel density estimates of each species' annual activity curves). These two metrics capture different aspects of temporal overlap: escape probability is directional and onset-anchored, measuring the pollination-relevant quantity of whether pollinator activity coincides with the flower's receptive window; OVL is symmetric and whole-distribution, measuring overall activity-curve similarity. In this system they diverge — OVL stable, escape probability collapsing — which we interpret as evidence that the mismatch is specifically eroding the onset-anchored pollination window rather than reflecting a general shift in both species' distributions.

The estimator works best when annual observations are numerous enough to characterize the leading edge of seasonal activity but not assumed to be exhaustive. It fails or becomes unstable when there are fewer than 30 records per species-year, when phenophase annotations are absent and non-flowering observations dominate, or when observer effort is strongly correlated with detection timing. The North American plant-side bias illustrates why observer-effort diagnostics must be part of the method, not an afterthought. Independent plant-onset validation using USDA/NASS bloom reports or state extension records is the highest-priority next step.

The robustness battery is meaningful only when enough valid annual PMI values exist. Three to six years can support exploratory slopes but not confirmatory inference. Seven or more valid years allow stronger screening, but decade-length series still cannot support formal climate attribution without additional evidence. The economic model is also deliberately conservative in interpretation: it estimates exposure under severity assumptions, not realized loss. It can overestimate risk when management adaptation offsets temporal mismatch and underestimate risk when nonlinear fruit-set losses occur before the severity score reaches its maximum.

Satellite phenology is the main constraint on global extension. Temperate regions can often be handled with dynamic NDVI thresholds, but tropical and semi-arid systems may require curve-fitting, harmonic analysis, phenocams, or biome-specific calibration. South Asia is therefore not a failed biological case; it is a boundary condition showing where threshold-based MODIS phenology must be replaced by more flexible seasonal-trajectory extraction.

The bounding-box approach to regional definition is a practical simplification with known limitations. Bounding boxes do not capture crop distribution, habitat suitability, elevation gradients, land-cover heterogeneity, or observer-community structure within regions. The North American bounding box includes commercial blueberry-growing regions and non-production areas where GBIF records may reflect different cultivars, wild relatives, or ornamental plantings. Crop masks, ecoregion filters, or species distribution model outputs could improve biological precision in future implementations. The current bounding-box design was chosen for reproducibility and transferability: any user can define a bounding box without requiring crop-distribution GIS layers that may not be publicly available for all regions.

The ten-year study window from 2015 to 2025 is sufficient for trend detection under the six-framework battery but short for formal climate attribution. A pre-registered distributed-lag model (PMI regressed on concurrent and one-year-lagged spring temperature) was not significant (joint $F(2,6) = 3.01$, $p = 0.124$), and a formal climate-sufficiency variance decomposition attributes only 22.6% of PMI variance to component-level temperature sensitivities (77.4% excess). Together these results show that the series is too short for temperature-based causal attribution and that even if it were longer, temperature alone cannot account for the magnitude of observed mismatch variation. Annual re-execution of the pipeline as additional years accumulate will progressively reduce this uncertainty. By approximately 2030, with fifteen annual observations, the series will be long enough to support more formal attribution analysis.

The Differential Climate Sensitivity hypothesis

If the Differential Climate Sensitivity hypothesis is correct, systems with a large gap in thermal sensitivity between interacting species should show PMI increasing with warming, and the shared-cue correlation between species' interannual fluctuations should remain strong even after removing the March-April temperature anomaly as a predictor. We tested both predictions against the North American series. The sensitivity gap is 5.6 days per °C (Vaccinium −5.2, Bombus +0.4), the PMI trend is positive and significant, and the shared-cue test confirms strong co-movement (detrended $r = 0.774$, $p = 0.009$). However, the climate-sufficiency test shows that temperature alone accounts for only 22.6% of PMI variance — a partial rejection that refines rather than refutes the hypothesis. The shared cue exists, but it is not primarily the March-April temperature anomaly. We therefore propose a refined prediction: the shared cue is an integrated seasonal signal — accumulated degree-days, soil temperature, or snowmelt timing — that neither species' single-month thermal sensitivity captures, and this cue should be recoverable from multi-variable climate models applied to longer series. This refined prediction is falsifiable: systems with 15+ years of concurrent PMI and multi-variable climate data should be able to identify the specific environmental covariate that drives the shared-cue correlation, or reject the hypothesis if no such covariate emerges.

The Mismatch Ratchet hypothesis

If the Mismatch Ratchet hypothesis is correct, systems with a large differential climate sensitivity (>3 days per °C gap between interactants) should show monotonically increasing PMI with no observed reversals across all available multi-year phenological datasets, and should fail to show mean-reversion in formal time-series stationarity tests as series lengthen. The ratchet analogy is deliberate: mismatch can increase but not decrease under sustained directional environmental change because the less-sensitive interactant cannot compensate through faster advance. The North American blueberry system tests two predictions of this hypothesis. First, the PMI trend should be monotonic (supported: the Chow test finds no significant breakpoint, $F = 2.82$, $p = 0.14$; the CUSUM permutation test shows no structural shift, $p = 0.90$). Second, the series should not be mean-reverting (supported: ADF test $p = 0.903$, consistent with a unit-root process). The sensitivity gap of 5.6 days per °C exceeds the proposed 3 days per °C threshold. A decade of data from a single system is not sufficient to confirm a ratchet, but both predictions are met and the hypothesis is stated as a testable benchmark: any future multi-system analysis that finds mean-reverting PMI in a pair with a sensitivity gap exceeding 3 days per °C would falsify the ratchet claim for that class of systems. We invite researchers to test this against the other candidate systems identified here (Western European Apis-Brassica, South Asian Cuculus-proxy) as their series lengthen.

Implications for monitoring

The North American demonstration case is strong because the trend is monotonic, statistically robust, economically relevant, and explicitly caveated. The result survives six statistical frameworks, but the *Vaccinium* observer-effort bias means the estimate should be treated as a strong monitoring signal requiring prospective validation rather than a final causal estimate. This balance is the point of the method: PhenoSync is designed to identify where evidence is strong, where it is economically important but statistically uncertain, and where the data pipeline itself fails.

Annual re-execution requires downloading updated GBIF occurrence records for the focal taxa, requesting updated MODIS composites via AppEEARS, and updating the NOAA and FAOSTAT inputs. Total data acquisition time is approximately two to four hours for a researcher familiar with the APIs; processing time on a standard laptop is approximately four to six hours for the full pipeline. The marginal cost of each additional annual update therefore declines as the pipeline infrastructure is established, making PhenoSync suitable for integration into annual agricultural monitoring workflows.

The same framework can be rerun annually as GBIF, MODIS, NOAA, FAOSTAT, and validation data grow. Candidate systems for immediate application include European barn swallow and oak-caterpillar or vegetation green-up systems, European rapeseed-pollinator systems, and Indian mustard-honeybee systems as South Asian iNaturalist participation increases. Because the method is modular, regions can use occurrence-only PMI, satellite-proxy PMI, or crop-exposure PMI depending on available data.

The annually re-executable design is a deliberate feature. Each additional year of GBIF records reduces uncertainty in the trend estimate, and each additional validation dataset, including herbarium specimens, USDA/NASS bloom reports, state extension phenology records, or phenocam imagery, can progressively resolve the plant-side observer-effort caveat identified here. The pipeline is designed so that adding a new data source requires only adding a new input CSV and rerunning the relevant script; no structural changes to the workflow are needed. This makes PhenoSync suitable as an operational monitoring tool rather than a one-time analysis.

PhenoSync is designed for three user communities. Agricultural scientists and extension researchers can use it to flag emerging mismatch risks in their focal crop systems before field evidence accumulates. Conservation biologists can apply it to any interacting species pair in GBIF to screen for phenological decoupling without requiring long-term field infrastructure. Policy analysts and agricultural economists can use the PMI-severity-exposure output to prioritise adaptation investment across crop systems and regions.

For the North American blueberry system specifically, three adaptation pathways follow directly from the PMI signal. First, cultivar timing can be treated as a phenological management tool because highbush blueberry cultivars differ in bloom timing and duration. Second, managed pollinator deployment can be made adaptive rather than calendar-fixed, with staggered colony placement adjusted annually based on observed crop bloom and pollinator activity. Third, wild *Bombus* habitat management can increase early-season floral resources and improve nesting conditions, reducing the chance that delayed spring emergence compounds crop-pollination risk. Each of these responses is testable within the PhenoSync monitoring framework by tracking whether PMI narrows following implementation.

When a system crosses the confirmatory threshold under the six-framework battery, the appropriate response is not to report the trend as a finalised causal estimate but to initiate a second-phase validation protocol: contact state extension services or national phenology networks for independent onset records, commission targeted field surveys in the affected region, and commission economic analysis using crop-specific pollination response functions. PhenoSync is designed as a screening tool that triggers this second phase, not as a replacement for it.

Beyond the blueberry system, the most immediate application priority is the European honeybee and oilseed rape system. Germany, France, and the United Kingdom together account for a substantial fraction of European rapeseed production, and oilseed rape is strongly bee-dependent. The Western European *Apis mellifera* / *Brassica napus* system showed a positive but non-significant trend in the current analysis, placing it in the watch-and-validate category, but the modeled economic exposure already exceeded one billion USD in 2024 under the severity framework. This combination of high exposure and uncertain biological trend is precisely the situation where annual monitoring adds the most value: it either confirms that the trend is strengthening toward the confirmatory threshold or provides evidence that the system is stable, both of which are actionable management signals.

The South Asian cuckoo-vegetation system represents a different kind of priority. The *Cuculus canorus* GBIF record set is large enough for inference, but satellite phenology extraction failed because the NDVI time series did not produce a stable single-threshold green-up signal. This is a solvable technical problem:

double-logistic or harmonic regression approaches to NDVI seasonal decomposition have been validated in multiple remote-sensing settings and can be substituted for the threshold-crossing algorithm in a future version. South Asia is therefore a high-priority next target not because of biological data limitations but because of a satellite processing limitation that has a known solution.

Post-hoc registration was completed on OSF before manuscript writing, but it is not prospective preregistration. Its value is transparency: it timestamps the finalized workflow, primary outcome, and robustness suite while future annual reruns provide the true prospective test.

Supplementary analyses

Supplementary materials (`supplementary.md`) include simulation performance tables, percentile sensitivity, jackknife, autocorrelation diagnostics, full multi-region trend results, observer-effort diagnostics, economic sensitivity, price and yield uncertainty, projections, power checks, MODIS validation, and herbarium validation outputs.

Conclusions

The systematic monitoring infrastructure now exists to answer, for the first time at global scale, whether the temporal contract between crops and their pollinators is holding — and the first decade of evidence from North America shows it is not.

PhenoSync introduces a reusable method for phenological mismatch monitoring from open data, built on three linked methodological contributions. The adaptive percentile estimator addresses the first-event problem in effort-heterogeneous presence-only occurrence records, outperforming the naive sample minimum by a factor of thirty in RMSE under high observer-effort bias in simulation. The six-framework robustness battery addresses the short-time-series problem by requiring convergence across distributional, non-parametric, permutation, influence, and autocorrelation diagnostics before a trend is classified as confirmatory, a standard we recommend for any phenological monitoring study using fewer than twenty annual observations. The PMI-severity exposure model addresses the translation problem by converting timing mismatch into a comparable economic exposure indicator applicable to any crop system with published pollination-dependence coefficients and FAOSTAT coverage.

Applied to the North American highbush blueberry system, the method produces three primary empirical findings that transcend the methodological contribution, each of which survives pre-specified FDR correction within a three-test primary family (Bonferroni threshold 0.0167). First, mismatch is driven asymmetrically: *Vaccinium* advances at 5.03 days per year, *Bombus* at 0.46 days per year, and the slope difference is significant ($Z = 2.98$, adjusted $p = 0.004$). Second, the onset-anchored escape probability at a 21-day window declined by 99.1% over nine years (adjusted $p = 0.004$), indicating that the temporal window for pollination is specifically narrowing even though whole-season distributional overlap remains stable. Third, the observed PMI trend of 4.57 days per year is 4.7× larger than temperature-driven differential sensitivity alone can explain, with a formal climate-sufficiency test attributing only 22.6% of PMI variance to component-level temperature responses — a Shiller-style demonstration that mismatch is too volatile for the standard climate-forcing framework. A nine-figure economic exposure estimate ($723 million in 2024 under conservative dependence assumptions) is robust to uncertainty in pollination-dependence coefficients and FAOSTAT input variation.

Secondary systems ranging from European rapeseed to South Asian cuckoo-vegetation proxies show watch-list and data-limited outcomes that are informative rather than negative: they identify where economic exposure is high but biological evidence is uncertain, and where the limiting factor is satellite phenology rather than occurrence data. Distinguishing supported signals from watchlist systems from methodological boundary conditions is the first step toward prioritising monitoring investment and adaptation planning.

If the pattern observed here — a fast-advancing crop paired with a thermally unresponsive pollinator — proves common across other insect-pollinated systems, it implies that climate adaptation strategies focused on crop breeding for earlier bloom are actively widening, not narrowing, the pollination gap. Highbush blueberry breeders have selected for earlier-flowering cultivars to escape summer heat stress and extend the harvest window, but the results indicate that this agronomic optimization has an ecological cost: it advances the crop into a seasonal period where pollinator activity is still sparse. If the plant's thermal sensitivity exceeds its pollinator's across multiple systems, breeding programmes currently optimising for early flowering as a heat-avoidance strategy are unintentionally eroding the temporal overlap on which their yield gains depend. This unintended consequence warrants urgent attention from breeders and indicates that phenological compatibility with pollinators should be an explicit criterion in cultivar development. PhenoSync provides the monitoring framework to detect this pattern wherever it emerges: the pipeline is openly documented, annually re-executable, and immediately applicable to any GBIF-covered species pair with sufficient date-stamped occurrence records. As citizen science participation grows and validation datasets expand, the uncertainty in all regional estimates will decline and the framework will become progressively more powerful without requiring structural changes. The pipeline is designed to improve with collective use.

Figures

Main-text figures (strongest results only)

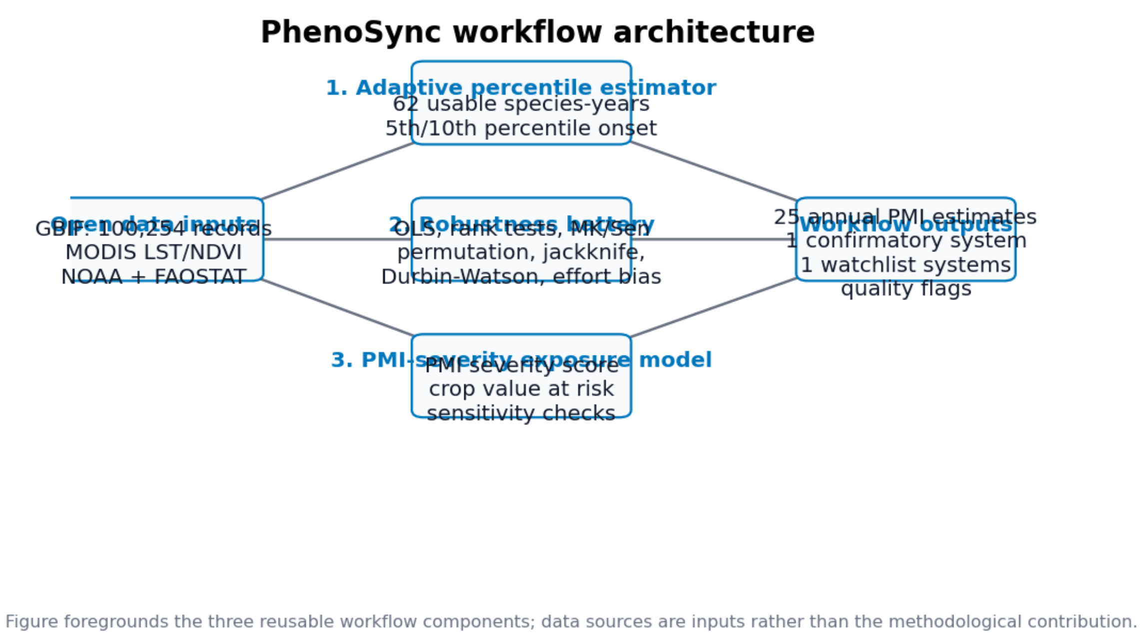

Figure 1: Figure 1

Figure 1. PhenoSync workflow showing the three methodological components. The workflow links the adaptive percentile estimator, robustness battery, and PMI-severity exposure model to their data inputs and outputs. Input data sources are shown left; methodological steps centre; outputs right. The pipeline is modular: economic exposure estimation requires FAOSTAT inputs but occurrence-only PMI can be computed

from GBIF records alone.

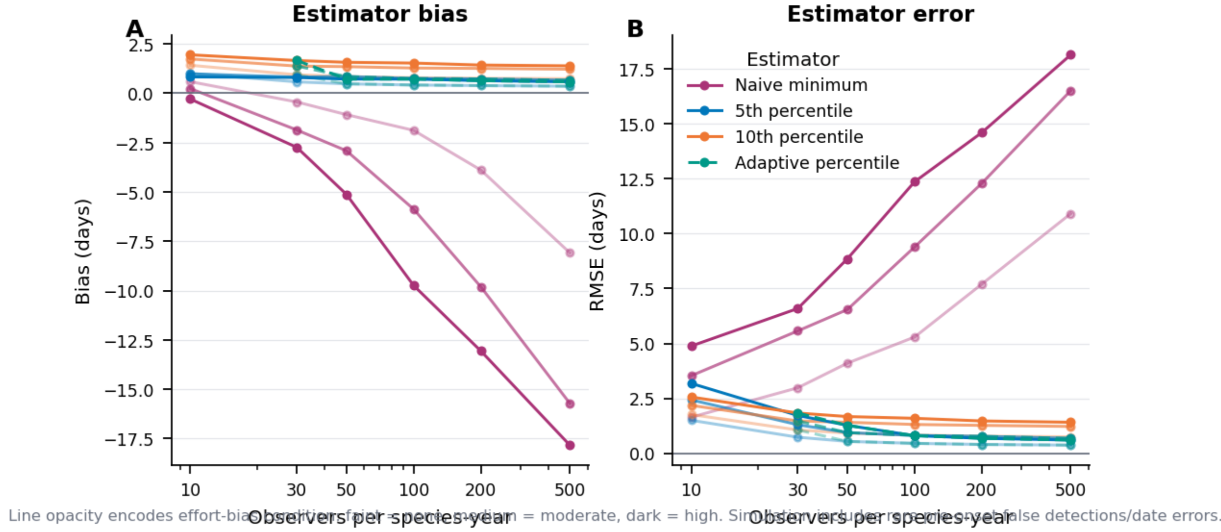

Figure 2: Figure 2

Figure 2. Simulation benchmark comparing naive sample minimum, fixed 5th percentile, fixed 10th percentile, and adaptive percentile estimators. (A) Estimator bias relative to true onset DOY = 100. (B) Root-mean-square error. Under high effort bias at 500 observers, RMSE was 18.14 days for the naive minimum versus 0.60 days for the adaptive estimator. Results are averaged over 500 iterations per condition.

Figure 3. Asymmetric phenological advance and overlap collapse in the North American highbush blueberry system, 2015-2025. Left panel: annual PMI (GBIF-derived solid blue, MODIS proxy dashed orange) with OLS trend (black line, slope = 4.57 days/year, 95% CI = [3.12, 6.02] days/year, $R^2 = 0.83$, $p = 0.0003$). Durbin-Watson = 2.45, indicating no substantial residual autocorrelation. Right panel: onset-anchored escape probability at W = 21 days (Bombus records within three weeks of Vaccinium onset), declining from 1.5% in 2016 to 0.1% in 2024 — a relative collapse of 99.1% (slope = −0.0021/yr, $p = 0.0037$, adjusted $p = 0.010$). The two strongest empirical results — monotonic mismatch accumulation and narrowing pollination window — are shown together as the study's primary visual summary.

Supplementary figures:

- Figure S1: Multi-region PMI trajectories (formerly main-text Figure 4)
- Figure S2: Annual GBIF record counts by taxon
- Figure S3: Bias-corrected PMI trend under observer-effort sensitivity analysis
- Figure S4: Stress-test extrapolations (formerly main-text Figure 6)
- Figure S5: Focal-region map with PMI severity scores (formerly main-text Figure 7)
- Figure S6: Distributed-lag climate model (formerly main-text Figure 5)
- Figure S7: Full robustness-battery diagnostic panels

Full supplementary materials are provided in `supplementary.md` to keep the main text focused on the two confirmatory primary results.

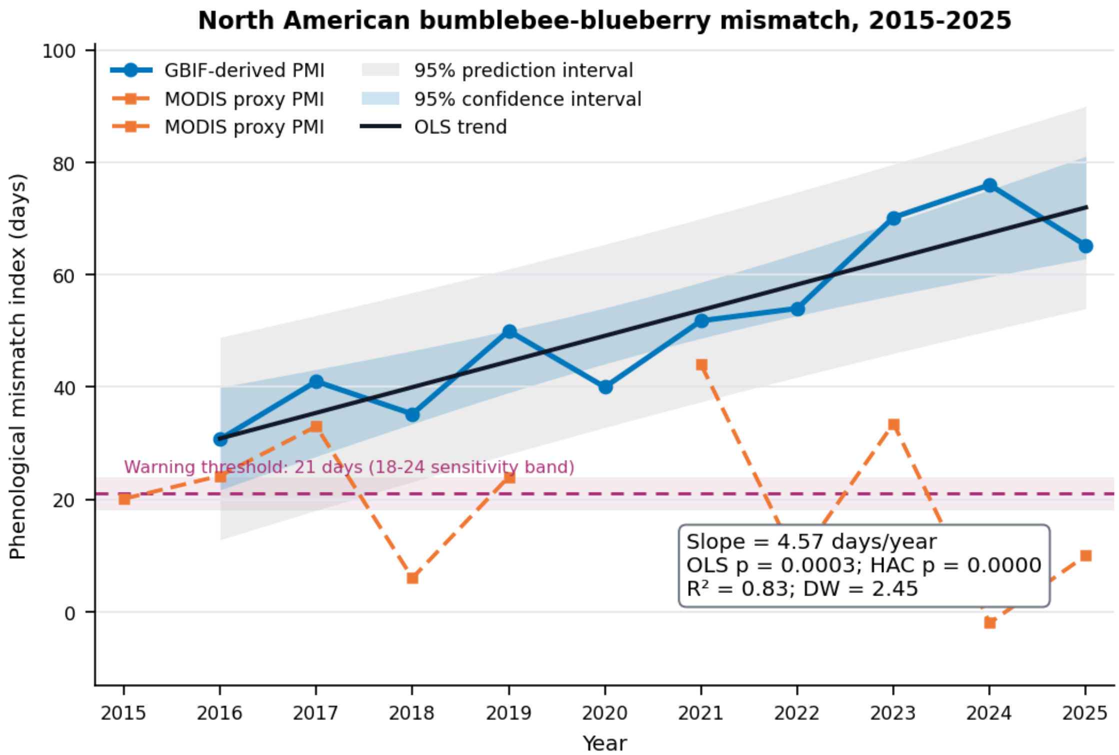

Figure 3: Figure 3

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